International Journal of Applied Earth Observations and Geoinformation 112 (2022) 102969

Contents lists available at ScienceDirect

International Journal of Applied Earth Observations and Geoinformation

journal homepage: www.elsevier.com/locate/jag

Exploring the capability of Gaofen-5 hyperspectral data for assessing soil salinity risks

Xiangyu Ge [a,b,c], Jianli Ding [a,b,c,*], Dexiong Teng [d], Boqiang Xie [a,b,c], Xianlong Zhang [e], Jinjie Wang [a,b,c], Lijing Han [a,b,c], Qingling Bao [a,b,c], Jingzhe Wang [f,g]

[a] College of Geography and Remote Sensing Science, Xinjiang University, Urumqi 800017, China
[b] Xinjiang Key Laboratory of Oasis Ecology, Xinjiang University, Urumqi 830017, China
[c] Key Laboratory of Smart City and Environment Modelling of Higher Education Institute, Xinjiang University, Urumqi 830017, China
[d] Institute of Applied Ecology, Chinese Academy of Sciences, Shenyang 110164, China
[e] School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China
[f] School of Artificial Intelligence, Shenzhen Polytechnic, Shenzhen 518055, China
[g] State Key Laboratory of Resources and Environmental Information System, Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, Beijing 100101, China

ARTICLE INFO

Keywords:
Gaofen-5
Soil salinization
Fractional order derivative
Machine learning
Digital soil mapping

ABSTRACT

Soil salinization has hampered the achievement of sustainable development goals (SDGs) in many countries worldwide. Several countries have recently launched hyperspectral remote sensing satellites, opening new avenues for accurate soil-salinity monitoring. Among them, Gaofen-5 (GF-5) from China has a high comprehensive performance, including a spectral resolution of 5 nm, 330 bands, and signal-to-noise ratio of 700. However, the potential of GF-5 for estimating soil salinity is not well understood. In this study, we proposed a strategy that includes bootstrap methods, fractional order derivative (FOD) techniques and decision-level fusion models to exploit the soil salinity diagnostic information and reduce estimation uncertainty in the Ebinur Lake oasis in northwestern China. The results showed that the GF-5 data were suitable for assessing soil salinity. The FOD technique enhanced the correlation between soil salinity and spectra, identified more diagnostic bands, improved the accuracy of soil salinity estimation, and reduced model uncertainty. The low-order FOD outperformed the high-order FOD. The spectra processed by the 0.9 order derivative were the most correlated with soil salinity ($r = -0.76$). The model driven by the 0.8 order derivative produced the optimal estimated model ($R^2 = 0.95$, root mean square error (RMSE) = 3.20 dS m^{-1} and a ratio of performance to interquartile distance (RPIQ) = 5.96). The model driven by the 0.8 order derivative had less uncertainty than the models based on the original and integer-order derivative (first- and second- derivatives) spectra. This study provides a reference for estimating soil salinity from GF-5 data using the proposed framework with low uncertainty and high accuracy. GF-5 data have great potential for assessing environmental problems and facilitating further SDGs.

1. Introduction

Soil salinization is one that causes land degradation worldwide. More than 100 countries and no less than 10 million km^2 of land suffer from soil salinity stress (Montanarella et al., 2015; Setia et al., 2013). This condition is a significant hindrance to progress toward the sustainable development goals (SDGs) worldwide (Erdogan et al., 2021; Mukhopadhyay et al., 2021; Singh, 2022; Tu et al., 2019). The continuous accumulation of soil salinity, which impedes plant extraction of nutrients owing to osmotic stress, reduces food production and ecological deterioration (Hassani et al., 2021; Ivushkin et al., 2019). Moreover, soil carbon sequestration is vulnerable to soil salinization. High soil salinity reduces vegetation productivity and decomposition (Zhao et al., 2018). Soil salinity can affect 50 % of croplands in the future, resulting in catastrophic economic losses (Corwin, 2021; Foley Jonathan et al., 2005). Furthermore, increased soil salinization globally depletes soil organic carbon by at least 6 Pg (Setia et al., 2013).

The use of remote sensing to monitor soil salinity has demonstrated to be successful owing to the advantages of rapid and large-scale monitoring over traditional sampling (Ge et al., 2022; Peng et al.,

* Corresponding author at: College of Geography and Remote Sensing Science, Xinjiang University, No. 777 Huarui Street, Urumqi, Xinjiang 830017, China.
E-mail address: watarid@xju.edu.cn (J. Ding).

https://doi.org/10.1016/j.jag.2022.102969
Received 24 May 2022; Received in revised form 23 July 2022; Accepted 7 August 2022
Available online 13 August 2022
1569-8432/© 2022 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

2019). In general, spectral reflectance is proportionate to soil salinity (Metternicht and Zinck, 2003). In previous studies, multispectral remote sensing, including Landsat and Sentinel-2 satellite data, has been widely used for soil salinity assessment (Fan et al., 2016; Ge et al., 2022; Wang et al., 2021). However, multispectral satellites have wide spectral channels and few bands, resulting in poor spectral detail and a lack of bands that are sensitive to soil salinity. Hyperspectral remote sensing satellites offer new perspectives for solving these complications. Satellite-based hyperspectral remote sensing data, represented by Hyperion and HJ-1 data, were initially used to quantify soil salinization. (Hamzeh et al., 2013; Weng et al., 2010). However, the EO-1 satellite is currently out of service and HJ-1 data do not have a shortwave infrared band (SWIR) or finer spatial resolution (Bai et al., 2018a; Wang et al., 2020b). Although these hyperspectral satellites effectively monitor soil salinity, the accuracy of the models constructed from hyperspectral data is only comparable to that of Landsat data (Bannari et al., 2020; Meng et al., 2020; Shi et al., 2022; Weng et al., 2010). The cause of the low estimation accuracy is an insufficient signal-to-noise ratio (SNR). Therefore, a high-accuracy estimation of soil salinity requires finer spatial and spectral resolution and higher SNR. The Gaofen-5 (GF-5) satellite with the Advanced Hyper-Spectral Imager (AHSI) performs well as a hyperspectral satellite in orbit. Compared with other hyperspectral satellites still operation, GF-5, launched by China, has a finer spatial resolution (30 m), a spectral range covering the visible, near-infrared (NIR), SWIR (range of 400–2500 nm), a better spectral resolution (the optimal value is 5 nm), the largest number of spectral bands (330 bands), a better SNR (maximum value of SNR is 700), and a revisit cycle of five days (Liu et al., 2019; Zhang et al., 2022). The GF-5 data were preliminary attempts to estimate other soil properties (Meng et al., 2020; Zhang et al., 2022). However, the capabilities of hyperspectral data from GF-5 still need to be explored further.

Hyperspectral data requires preprocessing to reduce interference from environmental factors. Spectral preprocessing is a prerequisite for exploiting the response mechanisms of spectra and soils (Abdul Munnaf et al., 2019). In general, the spectral derivative is applied to reduce background noise and eliminate the inter-spectral overlap effect (Appeltans et al., 2021; Demetriades-Shah et al., 1990; Wang et al., 2015). However, because the integer-order derivatives (first- and second-order derivatives) detect only the gradient of the spectrum, the subtle correlation between the spectrum and soil properties may be overlooked. Although the integer-order effectively enhances the spectral information, it reduces the spectral SNR owing to high-frequency noise and derivative increments (Hong et al., 2020). Fractional order derivatives (FOD) develop the concept of integer-order derivatives (Wang et al., 2018a). The FOD is continuously interpolated among integer-order derivatives to preserve the subtle information in the spectrum. The asymptotic derivation of the FOD overcomes the problem of integer-order processing diminishing the spectral SNR in hyperspectral remote sensing images owing to the high level of noise in these images (Hong et al., 2020; Wang et al., 2018b). FOD technology can extract useful spectral information on soil properties. Previous studies focused on spectral data (Ge et al., 2021; Hong et al., 2019); however, the FOD technique has rarely been applied to satellite-based hyperspectral data. Therefore, the FOD technique needs to be improved to process hyperspectral images with spectral and image characteristics and evaluate its effectiveness.

To fill these gaps, two important questions need to be addressed. Is GF-5 data appropriate for estimating the soil salinity? What does the FOD technique produce when it is applied to hyperspectral data? Specifically, the objectives are: (1) to explore the relationship between the spectral information of GF-5 and soil salinity, (2) to analyze the effect of FOD pretreatment on soil salinity estimation, and (3) to assess the uncertainty of the GF-5 original spectrum and FOD spectrum in estimating soil salinity.

2. Materials and methods

The workflow included data acquisition, pretreatment with FOD, modeling strategy and uncertainty analysis (Fig. 1). The following sections provide further information.

2.1. Study area and ground measurements

The Ebinur Lake oasis (43°38′-45°52′N, 79°53′-85°02′E) is essential national nature reserve in Northwest China (Fig. 2). Evaporation (1500 mm per year) is much greater than precipitation (200 mm per year) in this area. The soil orders are primarily Entisols, Aridosols, and Mollisols according to the soil taxonomy of the United States Department of Agriculture (USDA). *Populus euphratica* and *Haloxylon ammodendron* are the dominant natural plants. *Gossypium* spp is the main artificial plant. Cotton was completed by seeding before April 25 (Feng et al., 2017). Water for human production and consumption competes with that for ecological maintenance. Soil salinization was severe.

The field survey was conducted between May 16 and May 22, 2019. Total of 74 topsoil (0–10 cm) samples were collected. In a 30 m × 30 m sampling quadrat, each soil sample was blended using a five-point composite sampling method (the center point and four corners of each sample quadrat) (Fig. 2b). A soil auger was used to collect soil samples in sealed bags, and a handheld GPS device (GPS accuracy of approximately 1 m) was used to record the soil location coordinates. Bare soil remained predominant in most of the study area during fieldwork. There was a plastic film covering the agricultural fields, and cotton did not fully emerge. In the laboratory, soil samples were air-dried and passed through a 2 mm soil sieve. The soil leachate was prepared at a ratio of 1:5 soil to water and filtered at an indoor temperature of 25 °C. Soil electrical conductivity (EC) was measured in the soil leachate using a WTW™ Multiparameter (Multi 3420 SetB, Germany).

2.2. Gaofen-5 data collection and processing

The hyperspectral data were captured by the Advanced Hyper-Spectral Imager (AHSI) of the Gaofen-5 satellite as a data source. GF-5 data were downloaded from the China National Space Administration Earth Observation and Data Center (https://www.cheosgrid.org.cn/). ASHI data (two scenes) were obtained on May 18, 2019. The basic information of the GF-5 ASHI data is summarized in Table 1.

AHSI data processing includes radiometric calibration, atmospheric correction, topographic correction, mosaicking, and band removal. The seamless mosaic method was utilized for the mosaicking of two images. Even though the ASHI data contained 330 bands, 25 of them (193–200 bands and 246–262 bands from SWIR) were removed owing to their poor detector capture signal and spectral reflectance less than or equal to zero. Finally, 305 valid bands were retained, with 1–150 bands from VNIR, 151–192, 201–245, and 263–330 bands from SWIR. The above process was implemented in ENVI 5.3 (Harris Geospatial Corporation, USA).

2.3. FOD

FOD extends integer-order derivatives. In general, first- and second-order derivatives are considered integer-order derivatives (Hong et al., 2019). The FOD theory in Euclidean space extends the order from integer to non-integer and is unified with integer-order derivatives in the definition (Pu et al., 2008). For example, first-order and the second-order derivatives can be derived from the fractional-order derivative. FOD is suitable for signal and dispersion processing (Pu et al., 2010). In practice, numerous studies have adopted the Grünwald–Letnikov (G-L) definitions to apply to fractal questions (Hong et al., 2019). The G-L method is concise as it is deduced from the definition of integer-order, thus averting the complex Cauchy equation compared to the other definitions.

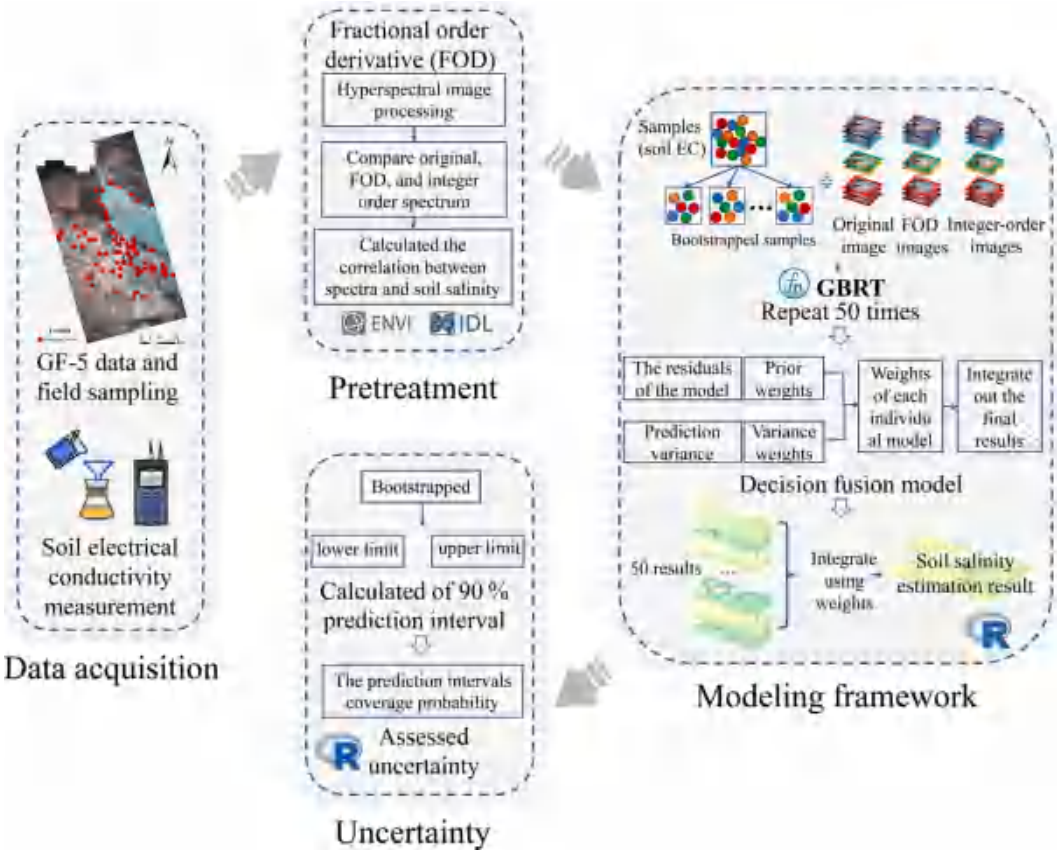

Fig. 1. The workflow of this study.

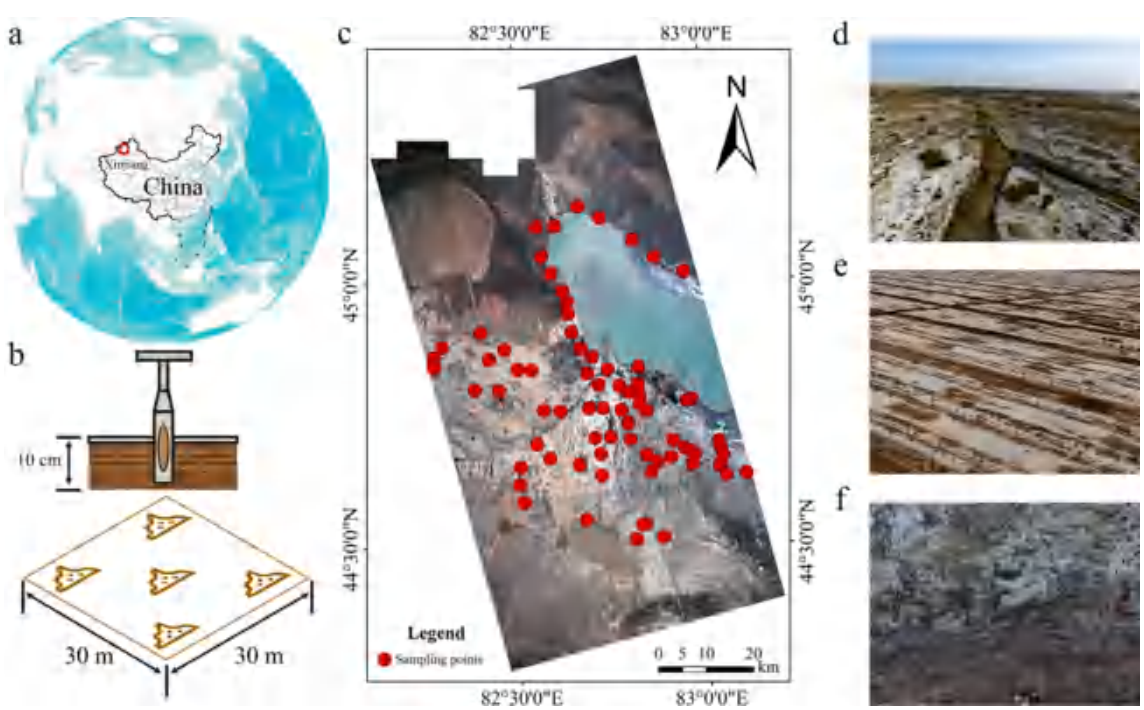

Fig. 2. The geographical location of Ebinur Lake and sampling points. (a) is location of study, (b) is sampling quadrat and sampling method, (c) is sampling points, (d) – (f) are landscape photograph of Ebinur Lake regions from UAV-based images (Photo by Xiangyu Ge used DJI Phantom 4).

Table 1
Main parameters of GF-5 ASHI.

Parameter	Characteristics
Spatial resolution (m)	30
Swath width (km)	60
Spectral range (nm)	390–2510
Spectral bands	330
Spectral resolution (nm)	5 (VNIR), 10 (SWIR)
SNR	700 (VNIR), 500 (SWIR)

VNIR, visible, near-infrared; SWIR, shortwave infrared; SNR, signal-to-noise ratio.

According to the theory of first-order derivatives, the formula for the function $f(x)$ is as follows:

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} \tag{1}$$

where h is the increment of the independent variable x.

The second order derivative of the function $f(x)$ is as follows:

$$f''(x) = \lim_{h\to 0}\frac{f'(x+h)-f'(x)}{h} = \lim_{h\to 0}\frac{f(x+2h)-2f(x+h)+f(x)}{h^2} \tag{2}$$

If the derivative order of function $f(x)$ is raised to a higher n order, then the nth order derivative of function $f(x)$ as follows:

$$f^{(n)}(x) = \lim_{h\to 0}\frac{1}{h^n}\sum_{m=0}^{n}(-1)^m\binom{n}{m}f(x-mh) \tag{3}$$

Here, replacing the binomial coefficients with the Gamma function and the n order expanding to non-integers, the G-L fractional order derivative formula can be obtained. The nth order of function $f(x)$ on interval $[a, t]$ is defined as follows:

$${}_aD_t^n f(x) = \lim_{h\to 0}\frac{1}{h^n}\sum_{m=0}^{\left[\frac{t-a}{h}\right]}(-1)^m\frac{\Gamma(n+1)}{m!\Gamma(n-m+1)}f(x-mh) \tag{4}$$

where h is an integral scale, $[\frac{t-a}{h}]$ is an integer part of $\frac{t-a}{h}$; and the Gamma function is $\Gamma(\tau) = (\tau-1)!$. Here, $n = 1$ expresses the first-order derivative and $n = 2$ denotes the second-order derivative. The formula for the nth order FOD of one-dimensional data at a continuous unit scale is as follows:

$$\begin{aligned}\frac{d^n f(x)}{dx^n} &\approx f(x) + (-n)f(x-1) \\ &+\frac{(-n)(-n+1)}{2}f(x-2) \\ &+\frac{(-n)(-n+1)(-n+2)}{6}f(x-3) \\ &+\cdots \\ &+\frac{\Gamma(-n+1)}{m!\Gamma(-n+m+1)}f(x-m)\end{aligned} \tag{5}$$

The two-dimensional image $f(x, y)$ is derived from Eq. (5) as follows:

$$\begin{aligned}\frac{d^n f(x,y)}{dx^n} &\approx f(x,y) + (-n)f(x-1,y) \\ &+\frac{(-n)(-n+1)}{2}f(x-2,y) \\ &+\frac{(-n)(-n+1)(-n+2)}{6}f(x-3,y) \\ &+\cdots \\ &+\frac{\Gamma(-n+1)}{m!\Gamma(-n+m+1)}f(x-m,y) \\ \frac{d^n f(x,y)}{dy^n} &\approx f(x,y) + (-n)f(x,y-1) \\ &+\frac{(-n)(-n+1)}{2}f(x,y-2) \\ &+\frac{(-n)(-n+1)(-n+2)}{6}f(x,y-3) \\ &+\cdots \\ &+\frac{\Gamma(-n+1)}{m!\Gamma(-n+m+1)}f(x,y-m)\end{aligned} \tag{6}$$

Considering the eight directions of the central pixel neighborhood, we superimpose the n-order fractional-order derivative 8-direction filter starting from the positive direction of the x-axis to obtain the FOD filter as follows:

$$\begin{bmatrix} a_m & & & a_m & & & a_m \\ & \ddots & & \vdots & & \cdots & \\ & & a_1 & a_1 & a_1 & & \\ a_m & \cdots & a_1 & & a_1 & \cdots & a_m \\ & & a_1 & a_1 & a_1 & & \\ & \cdots & & \vdots & & \ddots & \\ a_m & & & a_m & & & a_m \end{bmatrix} \tag{7}$$

where the respective coefficients are as follows:

$$\begin{cases} a_1 = -n \\ a_2 = \dfrac{(-n)(-n+1)}{2} \\ a_3 = \dfrac{(-n)(-n+1)(-n+2)}{3!} \\ \cdots \\ a_m = \dfrac{(-n)(-n+1)(-n+2)\cdots(-n+1-m)}{m!} \end{cases} \tag{8}$$

In this study, the filter size of the FOD used was 3×3, which was calculated in ENVI version 5.3 and IDL version 8.5.

2.4. Modeling strategies

In this study, the calibration and validation sets were randomly divided according to a 7:3 ratio. The dependent variable was the soil EC value and the independent variable was the spectrum under different FOD treatments. The modeling strategy combined the bootstrapped method and GBRT algorithm based on the original, FOD, and integer-order spectrum. This process was repeated 50 times to produce 50 soil-salinity maps. The 50 GBRT models were sub-models of the decision fusion model. Each model among the 50 iterations was assigned a weight, and the final result was integrated using the decision fusion model. As this modeling strategy is computationally intensive, modeling and prediction were performed using parallel computing.

2.4.1. Bootstrap

Bootstrapping is a nonparametric strategy for improving small-sample sizes and quantifying model uncertainty (Malone et al., 2011).

Bootstrapping is a resampling of the observed information, which strategy utilize random sampling as a replacement for generating the sample population described by a normal distribution (Wang et al., 2018b). The bootstrap approach has two advantages: (1) there is no need to model other assumptions and add new observations, and it is robust and efficient; and (2) by resampling, the sample reduction problem caused by cross-validation can be avoided. In this study, the bootstrapping strategy was combined with the GBRT algorithm to construct an estimated model (an introduction of the estimated model is detailed in the "GBRT" section), and then the model-derived soil salinity map. The bootstrapping method and modeling were repeated 50 times, resulting in 50 soil salinity maps. The bootstrapping strategy is also suitable for assessing the uncertainty of predicted model (Szatmári and Pásztor, 2019). The bootstrapping approach used the "BOOT" package in R 3.6.3.

2.4.2. GBRT

The GBRT model is an ensemble learning algorithm with a high generalization capability. The GBRT algorithm performs a first-order Taylor expansion on the loss function to reduce the residuals from the previous iteration for each iteration (Ridgeway, 2007). The value of the negative gradient direction of the loss function was used to approximate the residuals, and a regression tree was then fitted. Finally, the outcomes of all weak learners accumulate (Chen et al., 2019). Compared to the random forest model, GBRT offers three advantages: insensitivity to outliers, weight-based integration of weak learners, and simple parameter tweaking. In this study, bootstrapping of the optimal number of iterations in each model was obtained using 10-fold cross-validation. This method was performed utilizing the "gbm" package in R.

2.4.3. Decision fusion model

After constructing 50 soil salinity estimation models, it was necessary to integrate them. A decision fusion model was utilized to calculate the weights of 50 independent models to determine the effect of the integrated model (Peterson et al., 2019). Based on the estimated variance, this decision fusion model was constructed using a modified weighted-prior (MWP) method. This approach was a detailed introduction to previous research (Wang et al., 2020a). The introduction of the MWP-based decision fusion model can further reduce the bias in the regression results and improve the training approximation and generalization of the model. In this study, the decision fusion model was calculated using R.

2.5. Model evaluation

In this study, we utilized the coefficient of determination (R^2), root means square error (RMSE), and ratio of performance to interquartile distance (RPIQ) to evaluate the performance of all models. The formulas for these metrics can be found in previous studies (Ge et al., 2022). In general, an excellent model possesses an R^2 value closer to 1, an RMSE value closer to 0, and a higher RPIQ.

2.6. Uncertainty quantification

Prediction intervals are often considered credible methods for evaluating digital soil mapping. It is mainly expressed through the upper and lower prediction interval limits. The probability distribution for each predicted pixel was generated 50 times using a bootstrapping process. The mean of the 50 models and overall predicted variance were then calculated. We conducted the 90 % prediction interval and quantified the uncertainty of model prediction based on the specifications of the GlobalSoilMap project (Chen et al., 2022). The z value that corresponds to a 90 % probability was 1.645. Therefore, the upper and lower limits of 90 % prediction interval were calculated from the 50 bootstrapped iterations according to (Padarian et al., 2019) provided formula as follows:

$$\mathrm{PL} = \bar{x} \pm 1.645\sqrt{\delta^2 + AVMSE} \tag{9}$$

where PL is upper and lower limits of 90 % prediction interval, $\bar{x}$ denotes mean of the 50 models, the δ^2 indicates the variance of the 50 models, *AVMSE* expresses average mean square error of 50 validation sets.

The prediction intervals coverage probability (PICP) was used to determine whether the observations fall within the prediction interval of the respective estimation (Malone et al., 2011). PICP is the probability that all observation fit their prediction limits. In this study, a 90 % probability was calculated. The formula of PICP is as follows:

$$\mathrm{PICP} = \frac{1}{N} count(lowerPL_i < obs_i < upperPL_i) \times 100 \tag{10}$$

where N denotes the total number of observations in the validation set. The closer the PICP at each confidence level is to $100(1-\alpha)\%$, the better the model, and the closer their scatter is to the 1:1 line.

3. Results

3.1. Statistical description of soil samples

The distribution of soil EC in the soil samples is shown in Fig. 3. The mean soil EC was 14.44 dS m^{-1} and the range was 0.15–92.7 dS m^{-1}. The interquartile range (Q3-Q1), median and SD were 17.13 dS m^{-1}, 14.44 dS m^{-1} and 20.93 dS m^{-1}, respectively, which indicated soil EC was dispersion. The soil EC sample was positively skewed because the skewness (2.32) was greater than zero. Soil EC had a high relative variability. Moreover, the proportion of NS level was the largest, accounting for 35.14 % of the total soil salinity. Overall, soil salinity was asymmetrical and severe in the Ebinur Lake region.

3.2. Spectral characteristics of soil salinity levels

As soil salinity increased, soil reflectance increased, except in non-saline soils (Fig. 4). As non-saline soils came almost entirely from cotton fields covered with plastic mulch, the plastic mulch influenced the high reflectance. The characteristics of soil reflectance were similar for different salinity levels, with reflectance increasing steeply in the 390–960 nm and 1120–1470 nm. However, the soil spectra were relatively flat and increased in the visible range, while there was a notable difference in the 1123–1156 nm, 1452–1503 nm, and 1949–2109 nm for very slightly saline and slightly saline soil. This preliminary analysis provides a foundation for clarifying sensitive bands and estimating soil salinity.

3.3. FOD for hyperspectral data

The FOD spectra of all samples are showed in Fig. 5. In comparison, the FOD spectral disparities among the samples increased because the reflectance range became progressively wider. Spectral peaks became prominent near 960 nm, 1200 nm, 1460 nm, 1800 nm and 2000 nm. The results highlighted that FOD enhanced subtle differences in the spectra. When the order was greater than 1.0, the soil spectrum gradually lost its characteristics. Although the features around 1460 nm, 1800 nm and 2000 nm remained, the spectral information in the visible and NIR ranges was gradually eliminated.

The hyperspectral image after the FOD processing is shown in Fig. 6. At high-order FOD (order >1) as the order increases the image appears as a noisy pixel. The noisy pixels interfere with the image characteristics of the feature, which may be owing to the characteristics of the feature spectrum gradually disappear, and spectral noise appears. In comparison, the lower-order FOD outperformed others. The results indicated that lower-order FOD was more effective at emphasizing the characteristics and guaranteeing that the images were not distorted compared to the first-order and second-order derivatives.

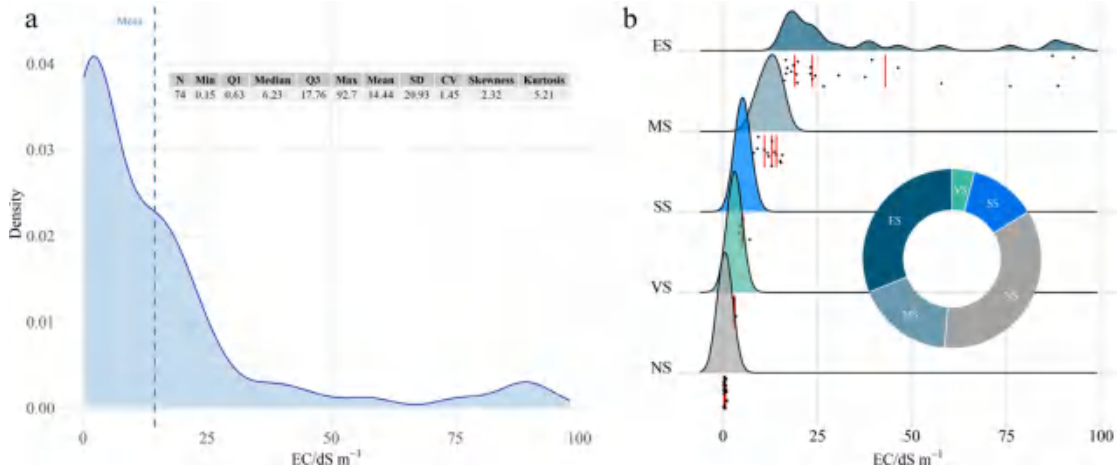

Fig. 3. The statistical analysis of soil EC. (a) is density distribution and descriptive statistics, (b) is sample distribution at different levels of soil salinity. The red lines are quartile. N is sample size; SD is standard deviation and CV means coefficient of variation. Min and Max express minimum and maximum, Q1 and Q3 denote lower quartile and upper quartile. NS: non-salinized, VS: very slightly salinized, SS: slightly salinized, MS: moderately salinized and ES: extremely salinized.

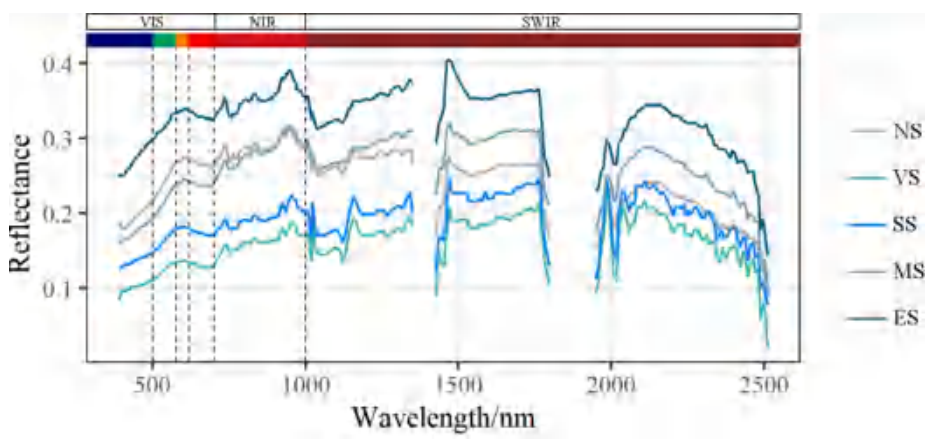

Fig. 4. Soil spectral characteristics of five saline grades. Shown here are the average spectra of soils at different soil salinity levels. NS: non-salinized, VS: very slightly salinized, SS: slightly salinized, MS: moderately salinized and ES: extremely salinized.

Fig. 7 portrays the correlation between the soil salinity and FOD spectra. The absolute values of the correlation coefficients (ACC) on the 0.1–1.6 order are higher than original counterparts. The ACC was 0.76 on the 0.9 order, which is 17 %, 0.8 %, and 26.5 % higher compared to the counterpart of the original spectrum, first- and second order derivative spectra, respectively. Furthermore, there were higher correlations on the 0.1–1.6 order near the wavelengths of 390–570 nm, 916–976 nm, 1427–1561 nm, 1772–1797 nm, 2151–2319 nm, 2345–2353 nm, 2412–2446 nm, and 2488 nm. However, the number of improved bands for high-order FOD was less than that for low-order FOD. The results highlighted that the FOD approach improved the correlation of the spectra with soil EC and tapped more sensitive bands.

3.4. Soil salinity estimation model evaluation

Fig. 8 illustrates the performance of the 50 iteration models. The 0.1–0.9 order, 1.1–1.3 order and 1.6–1.7 order models outperformed the original, first-order and second-order models. The 0.8 order model achieved the optimal performance. A small percentage of model accuracies were underachieved in all 50 iterations; nonetheless, they were superior to the other order models in this circumstance. These results indicate that the performance of model was ameliorated through FOD processing, but uncertainty remained.

The accuracy of the fusion model for each order is arrayed in Fig. 9. The 0.8-order model generated the optimal accuracy (validation set: $R^2 = 0.95$, RMSE = 3.20 dS m^{-1} and RPIQ = 5.96) in various modeling strategies. The R^2 of the 0.8 order model improved by 16.7 %, 8.2 %, and 22.4 %, respectively, when compared with the original, first-, and second-order derivative counterparts. The performance of the 0.9 order model was second best. However, not all FOD models ameliorated the accuracy of the estimated models. For example, the 1.7–1.9 order and second-order derivative models were less accurate than the original model. While the R^2 of the original spectral model was greater than that of the 1.7–1.9 order and the second-order derivative models, the RPIQ of the FOD and integer-order models were greater than those of the original spectral model. This indicated that the estimated soil EC of the FOD model was more consistent with the observed soil EC. In conclusion, the low-order model outperformed other models.

3.5. Soil salinity map and uncertainty quantification

To quantify the uncertainty of the optimal FOD model with the original spectral, integer-order model, Fig. 10 shows the spatial distribution of the lower and upper prediction interval limits, and the estimated soil EC. The spatial pattern of estimated soil salinity was similar except for the second-order derivative model. Stronger soil salinization was concentrated in the river inlet at the edge of the oasis at a lower elevation, whereas lower salinization occurred north of the lake and in

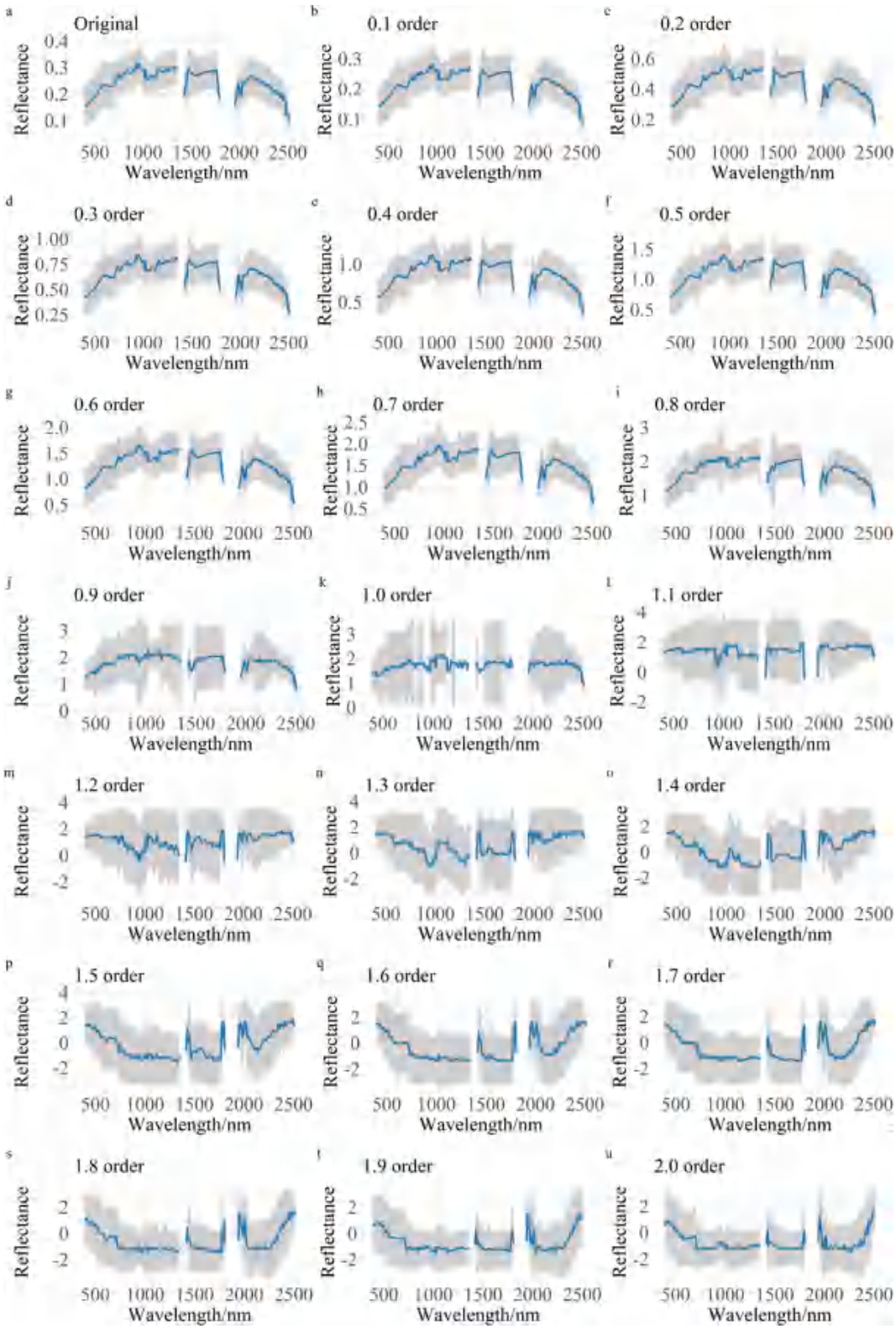

Fig. 5. Average spectrum during FOD processing. (a) is the original spectrum (order = 0). (b)-(t) are the 0.1–2.0 orders spectrum. The blue lines are average spectrum, the grey ranges are average spectrum plus or minus standard deviation.

the cotton croplands of the oasis. The mean value of each predicted soil EC map was within the mean value of the corresponding 90 % prediction interval limits. However, the second-order model had the most considerable uncertainty in this case, particularly for the estimation of the northern region.

To evaluate the sensitivity of the quantitative uncertainty, Fig. 11 compares the deviation of the PICP of these four models from the ideal confidence level. At the a PICP of 90 %, the models were close to the ideal confidence level. This result indicate that the defined 90 % prediction interval was suitable. The PICP of the 0.8 order model was

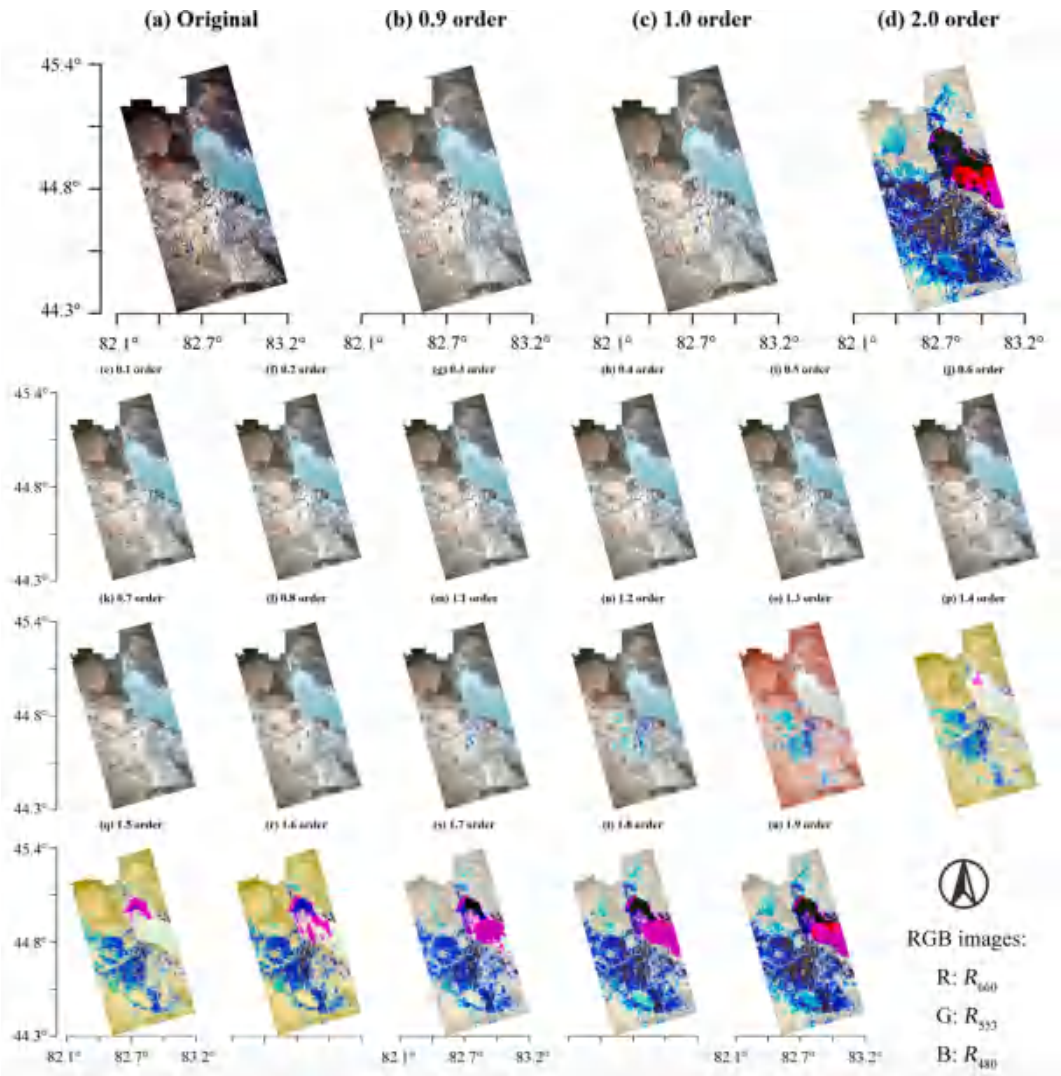

Fig. 6. The hyperspectral images during FOD processing. (a) is the original image, (b) is the 0.9 order deviation image, (c) is the first-order derivative, (d) is the second-order derivative. (c)-(u) are the 0.1–0.9 orders and 1.1–1.9 orders image.

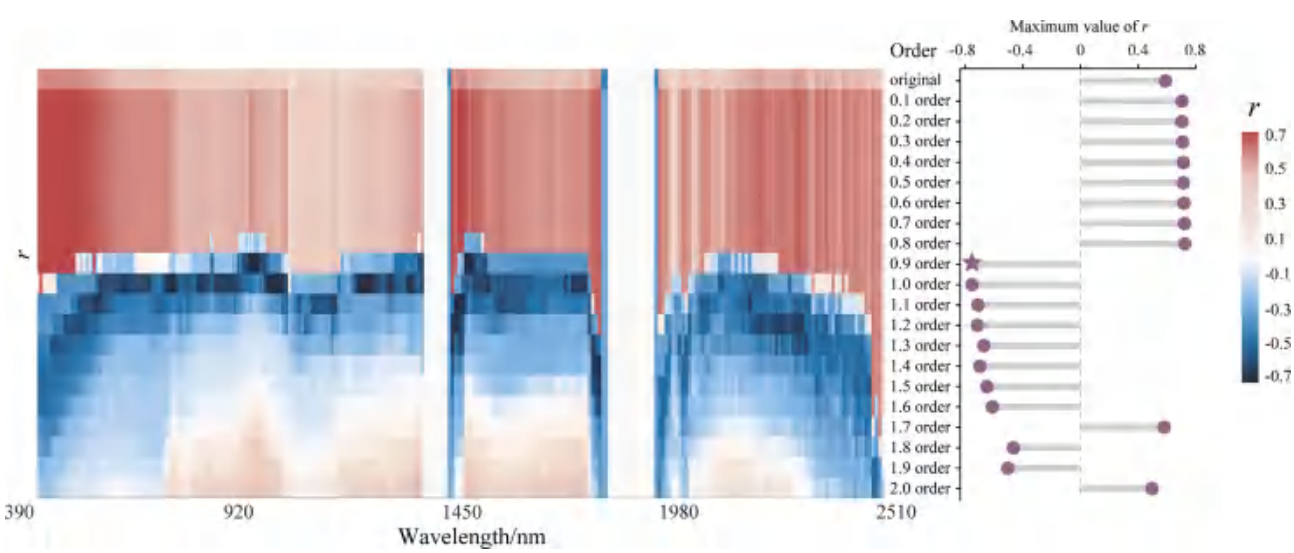

Fig. 7. The correlation of reflectance with soil EC at different orders.

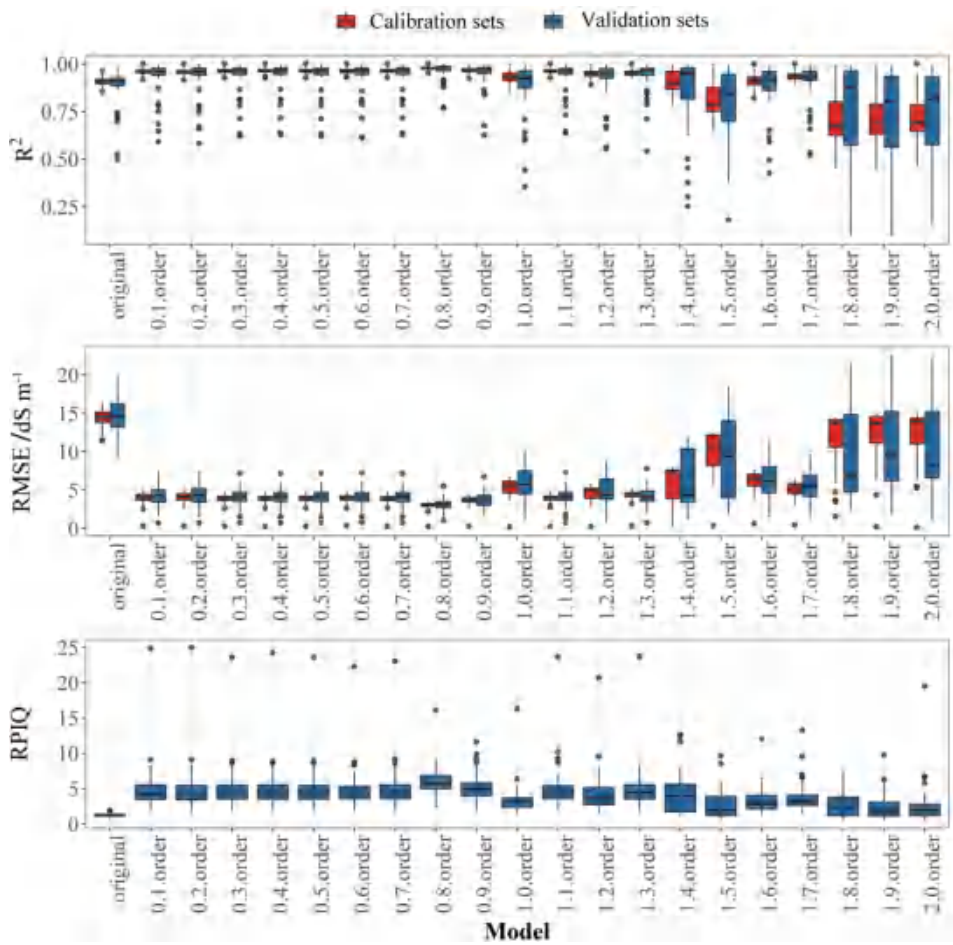

Fig. 8. The model performance statistics for 50 iterations. (a) is the results of R2, (b) is the results of RMSE, and (c) is the results of RPIQ.

closest to the 1:1 line. This study highlighted that the FOD model had lower uncertainty than the integer-order model. Overall, the 0.8-order model reduced the uncertainty in estimating soil salinity from the GF-5 ASHI data.

4. Discussion

4.1. Effect of GF-5 data for estimating soil salinity

GF-5 effectively estimated the soil salinity with high accuracy. This relies on the finer spectral resolution of the GF-5 data. Finer spectral information captured more details of the soil. For example, the differences in the ranges of 1123–1156 nm, 1452–1503 nm and 1949–2109 nm are more evident in this study. In contrast to previous studies, multispectral analysis only determined the reflectance pattern, and the differences among bands were not evident (Al-Ali et al., 2021; Bannari et al., 2020). In particular, direct monitoring of very slight and slight salinity using multispectral imagery is limited because different soil salinities have spectral characteristics (Fan et al., 2016; Masoud et al., 2019). Reviewing the *meta*-analysis of 57 papers on soil salinity estimation indicated that the average accuracy of multispectral images was 0.66–0.72 (Shi et al., 2022). In comparison, the accuracy of the original spectral model, in this case, was improved by 7–13 %. An accurate salinity estimation requires a large amount of detailed spectral information. In this region, NaCl dominates soil salinity, which has an impact on moisture absorption during the rainy season. (Wang et al., 2018a; Yang et al., 2021). Thus, in the original spectrum, 1452 nm was the most relevant to soil salinity (near the moisture absorption bands).

The high correlation between the spectra of GF-5 data and those of salt-affected soils measured in the laboratory was confirmed in a previous study (Wang et al., 2021). The soil reflectance obtained by GF-5 increased with increasing soil salinity and had a reflectance peak in the orange-yellow band. This result is consistent with the findings of this study. The bands sensitive to soil salinity were detected using Hyperion data, including bands at 437 nm, 548 nm, 569 nm, 630 nm, 640 nm, 813 nm, 1039 nm, 1487–1527 nm, 1971–1991 nm, 2032–2092 nm and 2163–2355 nm (Chang et al., 2008; Kumar et al., 2015; Weng et al., 2008), and HJ-1 hyperspectral data, including bands at 505.88 nm, 682.78 nm, 687.41 nm, 904.88 nm and 908.95 nm that are sensitive to soil salinity (Bai et al., 2018b; Li et al., 2015). These sensitive bands are comparable to those in this study, with the exception that more spectral bands in the original spectrum of GF-5 data were significantly correlated with soil salinity (Fig. 12). This outcome demonstrates the potential of GF-5 data for estimating the soil salinity. Furthermore, previous studies have shown that the accuracy of soil salinity estimation using hyperspectral data is similar to that using multispectral data (Davis et al., 2019; Shi et al., 2022). For example, the accuracy (R^2) of soil salinity estimation from HJ-1 data was 0.36–0.644 (Bai et al., 2018a; Li et al., 2015; Weng et al., 2008). Hyperion data were estimated with higher accuracy than HJ-1 ($R^2$of 0.63–0.78) (Kumar et al., 2015; Weng et al., 2010; Weng et al., 2008). In comparison, the original spectral model of GF-5 improved the estimation accuracy by 1–43 %. This likely because soil salinity mapping is affected by data noise. GF-5 had a higher SNR which is advantageous for soil salinity mapping.

In addition, plastic mulch acts as a confounding factor, boosting visible and NIR while decreasing SWIR reflectance (Han et al., 2022). This was consistent with the fact that non-salinized soils in this study had a higher reflectance. The soil salinity map from the original spectral

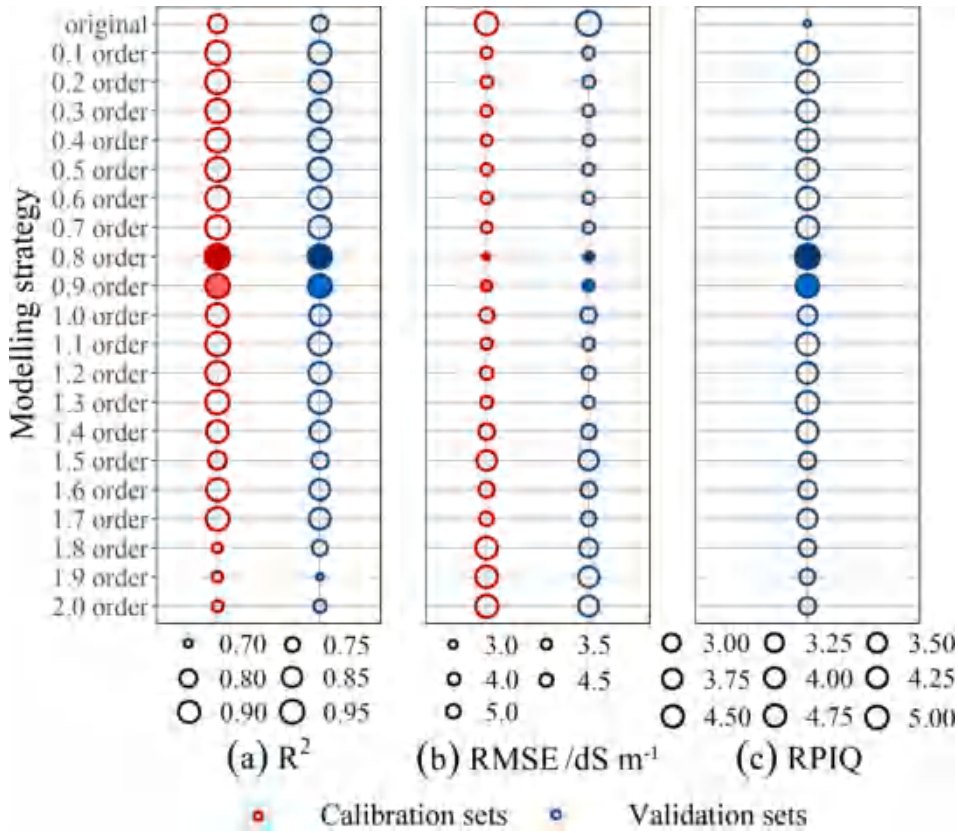

Fig. 9. The performance of various model. (a) is the results of R^2 values, (b) is the results of RMSE values, and (c) is the results of RPIQ values. The dark red or blue is performance of the optimal model, and the light red or blue is performance of suboptimal model.

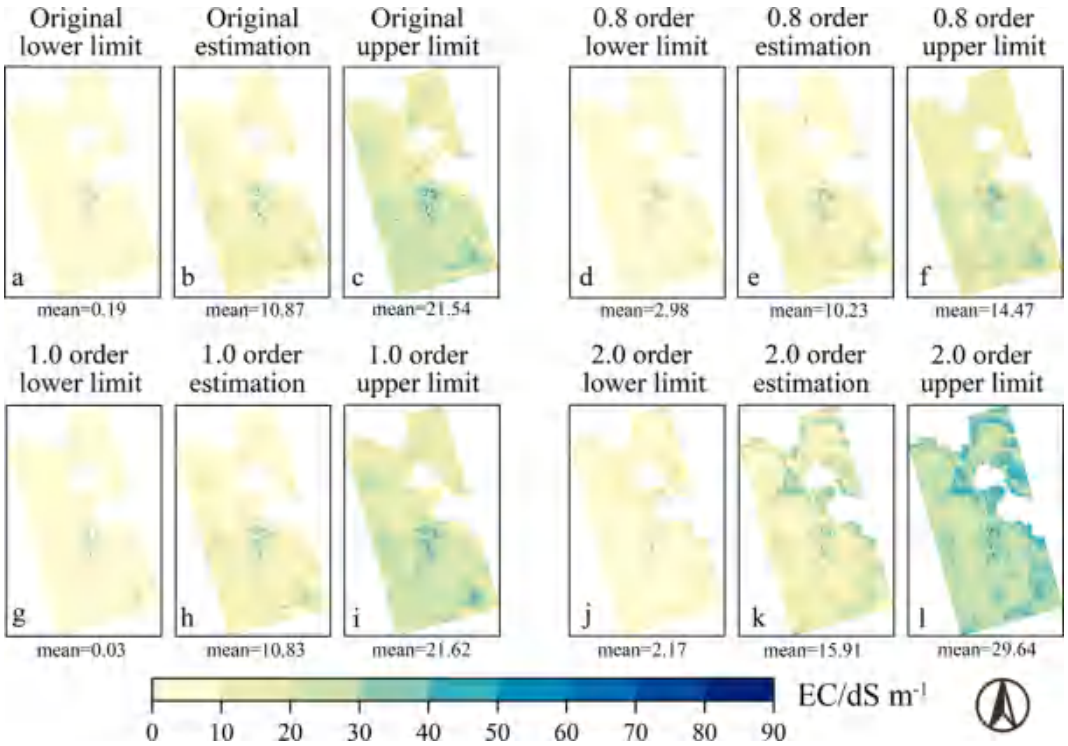

Fig. 10. The spatial distribution of soil salinity, lower and upper limit of the 90% prediction intervals. (a)-(c) are the results of original spectral model, (d)-(f) are the results of 0.8 order spectral model, (g)-(i) are the results of first-order spectral model, and (j)-(k) are the results of second-order spectral model.

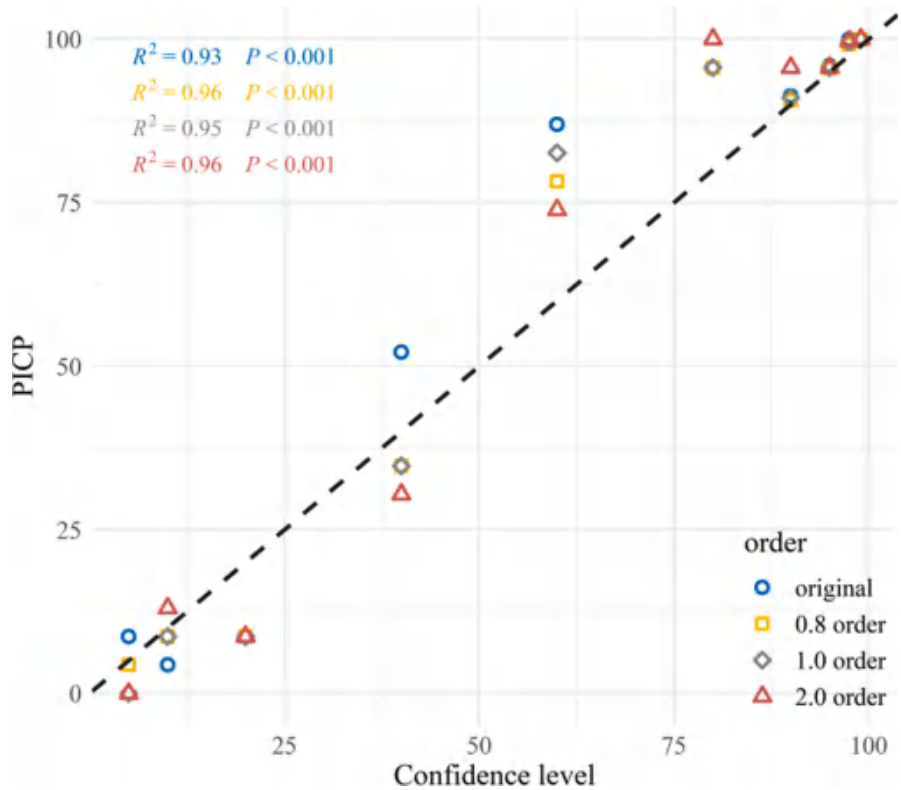

Fig. 11. PICP plot for various models on estimated soil EC.

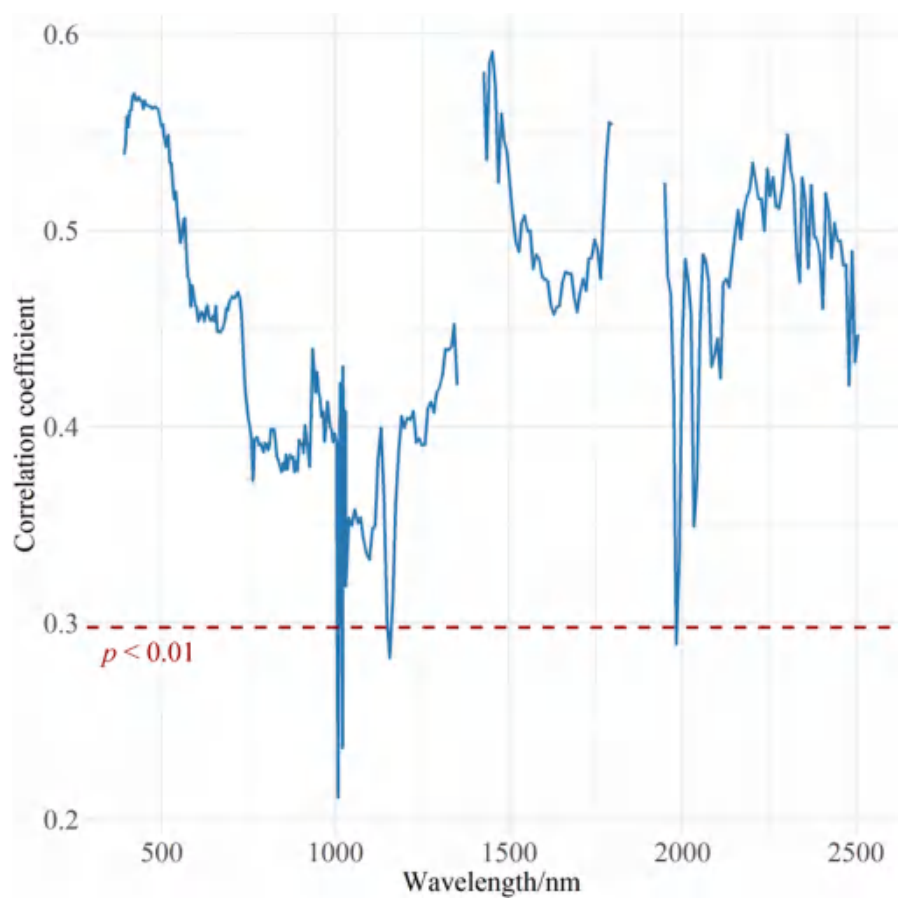

Fig. 12. Correlation of original spectra with measured soil EC. The red line is the critical value of the correlation coefficient ($p < 0.01$).

model demonstrated that cotton field had non-saline, very lightly saline, and lightly saline soils (Fig. 10) (Hu et al., 2021). The heavy soil salinity were found in the inlets of rivers with lower terrain after spring irrigation, which was consistent with earlier research (Wang et al., 2019). Overall, the GF-5 data are appropriate for assessing soil salinity at complicated surface circumstances.

4.2. The FOD technology advantages

Spectral derivatives are effective spectral processing methods (Wang et al., 2018b). By decreasing the noise and covariance concerns, spectral derivatives improve the correlation between the spectra and soil parameters (Munnaf et al., 2021). The FOD method emphasizes the soil spectrum details through minor step changes compared to the integer-orders. Spectral nuances are preserved because the fractional order is interpolated between the original, 1.0 and 2.0 order derivative spectrum. The FOD is essentially a generic slope of the spectral curve. Hence, the FOD retains the advantage of integer-order information while reducing the spectral distortion. In this study, the FOD method was improved to be suitable for two-dimensional images. Compared with integer-orders, the correlation between FOD spectra and soil salinity was improved by 2–17 %. Similar to previous studies, there are more salinity-responsive spectra (Wang et al., 2018a). Fig. 7 indicate that the best order with soil salinity is 0.9 order in this study, which was consistent with the study of ground-based spectra (Lao et al., 2021). Moreover, the FOD spectra correlated with the soil salts at approximately 916–976 nm. This is owing to the diagnostic absorption of halite (NaCl) in these bands (Howari et al., 2002). Through the ground-based spectrum, salt-sensitive bands at approximately 900 nm were identified (Pang et al., 2014). In comparison, satellite-based hyperspectral data are subjected to more significant atmospheric influence and background noise, obscuring diagnostic information in the raw spectra. However, not all FODs enhance the spectral information. Our results showed that a high-order FOD causes the spectra to lose soil spectral features and introduces noise to the images. (Ge et al., 2021) and (Hong et al., 2020) similarly found that high-order FOD was not conducive to hyperspectral processing. This likely owing to the increased noise peaks in the high-order FOD spectra, which interfere with the response information. Therefore, our findings confirmed that a low-order FOD significantly reduced interference and explored more effective diagnostic bands.

Furthermore, when compared to the original and integer-order spectra, the FOD spectra improved model accuracy and reduced model uncertainty. The optimal order model's R^2 was increased by 8.2–22.4 %. Interestingly, the order of the optimal model differed by 0.1 order from the most relevant order, which might suggest implementing a more detailed order, such as FOD at a 0.05 interval. This is because (Lao et al., 2021), in a study of ground-based spectra, the most relevant order for soil salinity and sodium ions was 0.9, whereas the most relevant order for chloride ions was 1.55 order. However, differences in salt ions dominate soil salinity in different regions, which could affect the results of using the FOD. The 0.8 order and first-order derivative models were closer to the 1:1 line when the uncertainty of the model was compared in Fig. 11. The spatial model also confirmed that the second-order derivative had low effectiveness. In previous soil property applications, the effect of FOD has not yet been fully understood. As uncertainty is necessary for estimating soil properties, assessing uncertainty enhances the interpretability of the model (Wadoux et al., 2020).

4.3. Research limitations and future work

Data, spectral covariates, and models are the most common sources of uncertainty (Wadoux et al., 2020). The bootstrapping method, FOD technique, and decision fusion model were used in this study to reduce uncertainty. Small sample size is a common problem in data-poor areas (Ge et al., 2022). Uncertainties in the spectral covariates may arise from the mixed spectra consisting of freshly germinated vegetation and bare soil. Another option is to apply an ensemble model such as a random forest (Douglas et al., 2018; Munnaf and Mouazen, 2022). However, the results from the individual model may be random, although ensemble learning models are suitable for soil property estimation. Therefore, combining multiple individual models may yield more stable results. The uncertainty of the results was low because each model was assigned a weight. Similar to this study, other research proposed super learning and a combined model (Abedi et al., 2021; Taghizadeh-Mehrjardi et al., 2021). The weights of the independent models in each modeling process were found to be inappropriate for each modeling result (Fig. 12a). The same conclusion was reached for the accuracy statistics (Fig. 9). Outlier models have the potential to degrade the results of the final model. These poor individual models should be assigned lower weights. Therefore, we proposed a strategy that overcame the small-sample problem and reduced the uncertainty.

We recommend avoiding single-modeling as much as possible in soil property mapping. However, this process requires a trade-off between the computational efficiency and model accuracy. The number of iterations, average mapping time, average R^2 value, and RMSE are shown in Fig. 13b. The average mapping time and average model accuracy increased with the number of iterations. However, the rate of increase in model accuracy slowed as the number of iterations increased. The mapping process required so much computational power that performing too many iterations would cause the computer's memory to run out. The number of iterations in previous studies was still set to 50 owing to the substantial computation required, even with parallel computing. In this study, the model accuracy gradually stabilized between 40 and 50 iterations; the average RMSE was in the range of 6.04 to 6.07. These findings suggest that a compromise between the accuracy and iterations is necessary. Recent developments in remote-sensing cloud-computing platforms, where the efficiency of computation has increased by a factor of 40 to 100 might bring new solutions to dense computing. In the future, higher accuracy soil property estimation may be greatly encouraged without worrying about the computational time constraints.

Machine learning, unlike geostatistical models, focuses on prediction accuracy. This necessitates quantifying the prediction uncertainty, but only 30 % of studies provide results (Wadoux et al., 2020). We suggest that combining soil genesis knowledge with uncertainty may provide more insight. For example, soil salinization is influenced by environmental factors. Although GF-5 provides a wealth of spectral data, it is insufficient for estimating the soil salinity at various depths. In addition, our research was a case study for evaluating GF-5 data to estimate soil salinity. The proposed method requires more data to validate the effect at other times and spaces. As more GF-5 satellites are launched, more data will be available for assessing soil salinity in other regions using the proposed strategy. The successive launches of hyperspectral satellites with finer spectral and spatial resolutions (such as EnMAP (Okujeni et al., 2015)) in various countries have increased the number of Earth observation opportunities.

5. Conclusions

In this study, we presented a lower uncertainty framework for estimating soil salinity using GF-5 hyperspectral data. GF-5 hyperspectral data can be used to estimate the soil salinity. FOD technology could improve the correlation between soil salinity and spectra, detect diagnostic feature bands, and lower model uncertainty. However, not all FODs perform well with hyperspectral data. The low-order FOD outperformed the high-order FOD. The highest correlation with soil salinity was found in the 0.9 order spectrum (r = -0.76). In comparison, the 0.8 order model has the highest accuracy (R^2 = 0.95, RMSE = 3.20 dS m^{-1} and RPIQ = 5.96) and the lowest uncertainty. We demonstrated that GF-5 data are suitable for estimating environmental problems under complex surface conditions, such as soil salinization. In the future, the bootstrap-FOD-fused decision model framework will help in low-uncertainty environmental monitoring.

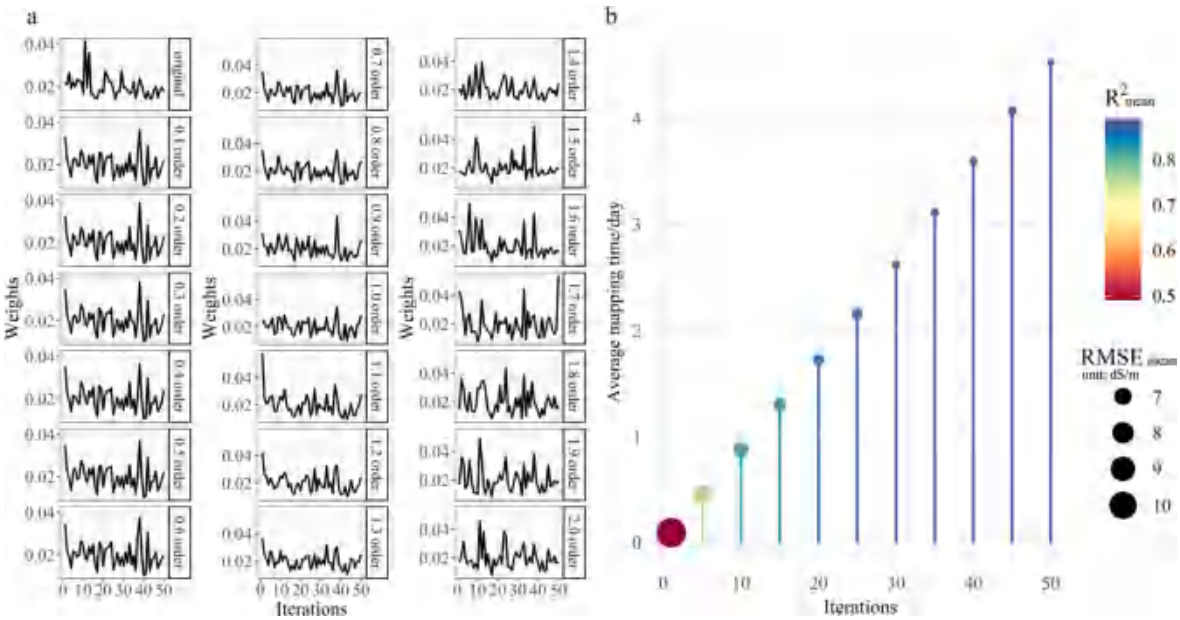

Fig. 13. The weights and efficiency of 50 iterations. (a) is the weight of 50 iterations on each model. (b) is the relationship between number of iterations, average model accuracy and average mapping time.

CRediT authorship contribution statement

Xiangyu Ge: Conceptualization, Methodology, Software, Visualization, Writing – original draft. **Jianli Ding:** Supervision, Writing – review & editing, Funding acquisition. **Dexiong Teng:** Software, Formal analysis. **Boqiang Xie:** Investigation, Validation. **Xianlong Zhang:** Methodology, Software. **Jinjie Wang:** Project administration, Resources. **Lijing Han:** Visualization, Data curation. **Qingling Bao:** Formal analysis, Visualization. **Jingzhe Wang:** Writing – review & editing, Methodology, Software.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Acknowledgements

We are sincerely grateful to the reviewers and editors for their constructive comments on the improvement of the manuscript. We thank the Shanghai Institute of Technical Physics of the Chinese Academy of Sciences for providing Gaofen-5 data to support our research. This study was supported by the Key Project of Natural Science Foundation of Xinjiang Uygur Autonomous Region (No.2021D01D06), the National Natural Science Foundation of China (No.41961059), the Key Laboratory of Spatial Data Mining & Information Sharing of Ministry of Education, Fuzhou University (No.2022LSDMIS05) and a grant from State Key Laboratory of Resources and Environmental Information System, China. The last, Ge Xiangyu is especially grateful to his parents for their continued support. He hopes his mother stays healthy after beating cancer.

References

Abdul Munnaf, M., Nawar, S., Mouazen, A.M., 2019. Estimation of Secondary Soil Properties by Fusion of Laboratory and On-Line Measured Vis–NIR Spectra. Remote Sensing 11, 2819.

Abedi, F., Amirian-Chakan, A., Faraji, M., Taghizadeh-Mehrjardi, R., Kerry, R., Razmjoue, D., Scholten, T., 2021. Salt dome related soil salinity in southern Iran: Prediction and mapping with averaging machine learning models. Land Degrad. Dev. 32 (3), 1540–1554.

Al-Ali, Z.M., Bannari, A., Rhinane, H., El-Battay, A., Shahid, S.A., Hameid, N., 2021. Validation and Comparison of Physical Models for Soil Salinity Mapping over an Arid Landscape Using Spectral Reflectance Measurements and Landsat-OLI Data. Remote Sensing 13, 494.

Appeltans, S., Pieters, J.G., Mouazen, A.M., 2021. Detection of Leek Rust Disease under Field Conditions Using Hyperspectral Proximal Sensing and Machine Learning. Remote Sensing 13, 1341.

Bai, L., Wang, C., Zang, S., Wu, C., Luo, J., Wu, Y., 2018a. Mapping Soil Alkalinity and Salinity in Northern Songnen Plain, China with the HJ-1 Hyperspectral Imager Data and Partial Least Squares Regression. Sensors 18, 3855.

Bai, L., Wang, C., Zang, S., Wu, C., Luo, J., Wu, Y., 2018b. Mapping Soil Alkalinity and Salinity in Northern Songnen Plain, China with the HJ-1 Hyperspectral Imager Data and Partial Least Squares Regression. Sensors 18 (11), 3855.

Bannari, A., Musa, N.H.M., Abuelgasim, A., El-Battay, A., 2020. Sentinel-MSI and Landsat-OLI Data Quality Characterization for High Temporal Frequency Monitoring of Soil Salinity Dynamic in an Arid Landscape. IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens. 13, 2434–2450.

Chang, C., Chen, X., Bao, A., Luo, G., 2008. Quantative Inversion based on Hyperspectral Data for Soil Salinization in the Ecological Corridor of Tarim River, Xinjiang, China, IGARSS 2008 - 2008 IEEE International Geoscience and Remote Sensing Symposium, pp. III - 1075-III - 1078.

Chen, S., Arrouays, D., Leatitia Mulder, V., Poggio, L., Minasny, B., Roudier, P., Libohova, Z., Lagacherie, P., Shi, Z., Hannam, J., Meersmans, J., Richer-de-Forges, A. C., Walter, C., 2022. Digital mapping of GlobalSoilMap soil properties at a broad scale: A review. Geoderma 409, 115567.

Chen, D., Chang, N., Xiao, J., Zhou, Q., Wu, W., 2019. Mapping dynamics of soil organic matter in croplands with MODIS data and machine learning algorithms. Sci. Total Environ. 669, 844–855.

Corwin, D.L., 2021. Climate change impacts on soil salinity in agricultural areas. Eur. J. Soil Sci. 72, 842–862.

Davis, E., Wang, C., Dow, K., 2019. Comparing Sentinel-2 MSI and Landsat 8 OLI in soil salinity detection: a case study of agricultural lands in coastal North Carolina. Int. J. Remote Sens. 40, 6134–6153.

Demetriades-Shah, T.H., Steven, M.D., Clark, J.A., 1990. High resolution derivative spectra in remote sensing. Remote Sens. Environ. 33, 55–64.

Douglas, R.K., Nawar, S., Alamar, M.C., Mouazen, A.M., Coulon, F., 2018. Rapid prediction of total petroleum hydrocarbons concentration in contaminated soil using vis-NIR spectroscopy and regression techniques. Sci. Total Environ. 616–617, 147–155.

Erdogan, H.E., Havlicek, E., Dazzi, C., Montanarella, L., Van Liedekerke, M., Vrščaj, B., Krasilnikov, P., Khasankhanova, G., Vargas, R., 2021. Soil conservation and sustainable development goals(SDGs) achievement in Europe and central Asia:

Which role for the European soil partnership? Int. Soil Water Conserv. Res. 9, 360–369.
Fan, X., Weng, Y., Tao, J., 2016. Towards decadal soil salinity mapping using Landsat time series data. Int. J. Appl. Earth Obs. Geoinf. 52, 32–41.
Feng, L., Dai, J., Tian, L., Zhang, H., Li, W., Dong, H., 2017. Review of the technology for high-yielding and efficient cotton cultivation in the northwest inland cotton-growing region of China. Field Crops Res. 208, 18–26.
Foley Jonathan, A., DeFries, R., Asner Gregory, P., Barford, C., Bonan, G., Carpenter Stephen, R., Chapin, F.S., Coe Michael, T., Daily Gretchen, C., Gibbs Holly, K., Helkowski Joseph, H., Holloway, T., Howard Erica, A., Kucharik Christopher, J., Monfreda, C., Patz Jonathan, A., Prentice, I.C., Ramankutty, N., Snyder Peter, K., 2005. Global Consequences of Land Use. Science 309, 570–574.
Ge, X., Ding, J., Jin, X., Wang, J., Chen, X., Li, X., Liu, J., Xie, B., 2021. Estimating Agricultural Soil Moisture Content through UAV-Based Hyperspectral Images in the Arid Region. Remote Sensing 13.
Ge, X., Ding, J., Teng, D., Wang, J., Huo, T., Jin, X., Wang, J., He, B., Han, L., 2022. Updated soil salinity with fine spatial resolution and high accuracy: The synergy of Sentinel-2 MSI, environmental covariates and hybrid machine learning approaches. Catena 212, 106054.
Hamzeh, S., Naseri, A.A., AlaviPanah, S.K., Mojaradi, B., Bartholomeus, H.M., Clevers, J. G.P.W., Behzad, M., 2013. Estimating salinity stress in sugarcane fields with spaceborne hyperspectral vegetation indices. Int. J. Appl. Earth Obs. Geoinf. 21, 282–290.
Han, L., Ding, J., Wang, J., Zhang, J., Xie, B., Hao, J., 2022. Monitoring Oasis Cotton Fields Expansion in Arid Zones Using the Google Earth Engine: A Case Study in the Ogan-Kucha River Oasis, Xinjiang, China. Remote Sensing 14, 225.
Hassani, A., Azapagic, A., Shokri, N., 2021. Global predictions of primary soil salinization under changing climate in the 21st century. Nat. Commun. 12, 6663.
Hong, Y., Liu, Y., Chen, Y., Liu, Y., Yu, L., Liu, Y., Cheng, H., 2019. Application of fractional-order derivative in the quantitative estimation of soil organic matter content through visible and near-infrared spectroscopy. Geoderma 337, 758–769.
Hong, Y., Guo, L., Chen, S., Linderman, M., Mouazen, A.M., Yu, L., Chen, Y., Liu, Y., Liu, Y., Cheng, H., Liu, Y., 2020. Exploring the potential of airborne hyperspectral image for estimating topsoil organic carbon: Effects of fractional-order derivative and optimal band combination algorithm. Geoderma 365, 114228.
Howari, F.M., Goodell, P.C., Miyamoto, S., 2002. Spectral Properties of Salt Crusts Formed on Saline Soils. J. Environ. Qual. 31, 1453–1461.
Hu, T., Hu, Y., Dong, J., Qiu, S., Peng, J., 2021. Integrating Sentinel-1/2 Data and Machine Learning to Map Cotton Fields in Northern Xinjiang, China. Remote Sensing 13, 4819.
Ivushkin, K., Bartholomeus, H., Bregt, A.K., Pulatov, A., Kempen, B., de Sousa, L., 2019. Global mapping of soil salinity change. Remote Sens. Environ. 231, 111260.
Kumar, S., Gautam, G., Saha, S.K., 2015. Hyperspectral remote sensing data derived spectral indices in characterizing salt-affected soils: a case study of Indo-Gangetic plains of India. Environ. Earth Sci. 73, 3299–3308.
Lao, C., Chen, J., Zhang, Z., Chen, Y., Ma, Y., Chen, H., Gu, X., Ning, J., Jin, J., Li, X., 2021. Predicting the contents of soil salt and major water-soluble ions with fractional-order derivative spectral indices and variable selection. Comput. Electron. Agric. 182, 106031.
Li, J., Pu, L., Zhu, M., Dai, X., Xu, Y., Chen, X., Zhang, L., Zhang, R., 2015. Monitoring soil salt content using HJ-1A hyperspectral data: A case study of coastal areas in Rudong County, Eastern China. Chinese Geogr. Sci. 25, 213–223.
Liu, Y.N., Sun, D.X., Hu, X.N., Ye, X., Li, Y.D., Liu, S.F., Cao, K.Q., Chai, M.Y., Zhou, W.Y. N., Zhang, J., Zhang, Y., Sun, W.W., Jiao, L.L., 2019. The Advanced Hyperspectral Imager: Aboard China's GaoFen-5 Satellite. IEEE Geosci. Remote Sens. Mag. 7, 23–32.
Malone, B.P., McBratney, A.B., Minasny, B., 2011. Empirical estimates of uncertainty for mapping continuous depth functions of soil attributes. Geoderma 160, 614–626.
Masoud, A.A., Koike, K., Atwia, M.G., El-Horiny, M.M., Gemail, K.S., 2019. Mapping soil salinity using spectral mixture analysis of landsat 8 OLI images to identify factors influencing salinization in an arid region. Int. J. Appl. Earth Obs. Geoinf. 83, 101944.
Meng, X., Bao, Y., Liu, J., Liu, H., Zhang, X., Zhang, Y., Wang, P., Tang, H., Kong, F., 2020. Regional soil organic carbon prediction model based on a discrete wavelet analysis of hyperspectral satellite data. Int. J. Appl. Earth Obs. Geoinf. 89, 102111.
Metternicht, G.I., Zinck, J.A., 2003. Remote sensing of soil salinity: potentials and constraints. Remote Sens. Environ. 85, 1–20.
Montanarella, L., Badraoui, M., Chude, V., Costa, I., Mamo, T., Yemefack, M., Aulang, M., Yagi, K., Hong, S.Y., Vijarnsorn, P.J.E.S.-L.C., 2015. Status of the world's soil resources: main report.
Mukhopadhyay, R., Sarkar, B., Jat, H.S., Sharma, P.C., Bolan, N.S., 2021. Soil salinity under climate change: Challenges for sustainable agriculture and food security. J. Environ. Manage. 280, 111736.
Munnaf, M.A., Guerrero, A., Nawar, S., Haesaert, G., Van Meirvenne, M., Mouazen, A.M., 2021. A combined data mining approach for on-line prediction of key soil quality indicators by Vis-NIR spectroscopy. Soil Tillage Res. 205, 104808.
Munnaf, M.A., Mouazen, A.M., 2022. Removal of external influences from on-line vis-NIR spectra for predicting soil organic carbon using machine learning. Catena 211, 106015.
Okujeni, A., van der Linden, S., Hostert, P., 2015. Extending the vegetation–impervious–soil model using simulated EnMAP data and machine learning. Remote Sens. Environ. 158, 69–80.
Padarian, J., Minasny, B., McBratney, A.B., 2019. Using deep learning for digital soil mapping. Soil 5, 79–89.
Pang, G., Wang, T., Liao, J., Li, S., 2014. Quantitative Model Based on Field-Derived Spectral Characteristics to Estimate Soil Salinity in Minqin County, China. Soil Sci. Soc. Am. J. 78, 546–555.
Peng, J., Biswas, A., Jiang, Q., Zhao, R., Hu, J., Hu, B., Shi, Z., 2019. Estimating soil salinity from remote sensing and terrain data in southern Xinjiang Province, China. Geoderma 337, 1309–1319.
Peterson, K.T., Sagan, V., Sidike, P., Hasenmueller, E.A., Sloan, J.J., Knouft, J.H., 2019. Machine Learning-Based Ensemble Prediction of Water-quality Variables Using Feature-level and Decision-level Fusion with Proximal Remote Sensing. Photogramm. Eng. Remote Sens. 85, 269–280.
Pu, Y., Wang, W., Zhou, J., Wang, Y., Jia, H., 2008. Fractional differential approach to detecting textural features of digital image and its fractional differential filter implementation. Sci. China Series F: Inform. Sci. 51, 1319–1339.
Pu, Y., Zhou, J., Yuan, X., 2010. Fractional Differential Mask: A Fractional Differential-Based Approach for Multiscale Texture Enhancement. IEEE Trans. Image Process. 19, 491–511.
Ridgeway, G.J.U., 2007. Generalized Boosted Models: A guide to the gbm package 1.
Setia, R., Gottschalk, P., Smith, P., Marschner, P., Baldock, J., Setia, D., Smith, J., 2013. Soil salinity decreases global soil organic carbon stocks. Sci. Total Environ. 465, 267–272.
Shi, H., Hellwich, O., Luo, G., Chen, C., He, H., Ochege, F.U., Voorde, T.V.D., Kurban, A., Maeyer, P.D., 2022. A Global Meta-Analysis of Soil Salinity Prediction Integrating Satellite Remote Sensing, Soil Sampling, and Machine Learning. IEEE Trans. Geosci. Remote Sens. 60, 1-15.
Singh, A., 2022. Soil salinity: A global threat to sustainable development. Soil Use Manag. 38, 39–67.
Szatmári, G., Pásztor, L., 2019. Comparison of various uncertainty modelling approaches based on geostatistics and machine learning algorithms. Geoderma 337, 1329–1340.
Taghizadeh-Mehrjardi, R., Hamzehpour, N., Hassanzadeh, M., Heung, B., Ghebleh Goydaragh, M., Schmidt, K., Scholten, T., 2021. Enhancing the accuracy of machine learning models using the super learner technique in digital soil mapping. Geoderma 399, 115108.
Tu, C., Suweis, S., D'Odorico, P., 2019. Impact of globalization on the resilience and sustainability of natural resources. Nat. Sustainability 2, 283–289.
Wadoux, A.M.J.C., Minasny, B., McBratney, A.B., 2020. Machine learning for digital soil mapping: Applications, challenges and suggested solutions. Earth Sci. Rev. 210, 103359.
Wang, J., Ding, J., Abulimiti, A., Cai, L., 2018a. Quantitative estimation of soil salinity by means of different modeling methods and visible-near infrared (VIS–NIR) spectroscopy, Ebinur Lake Wetland, Northwest China. PeerJ 6, e4703.
Wang, J., Ding, J., Yu, D., Ma, X., Zhang, Z., Ge, X., Teng, D., Li, X., Liang, J., Lizaga, I., Chen, X., Yuan, L., Guo, Y., 2019. Capability of Sentinel-2 MSI data for monitoring and mapping of soil salinity in dry and wet seasons in the Ebinur Lake region, Xinjiang, China. Geoderma 353, 172–187.
Wang, Z., Zhang, X., Zhang, F., Chan, N.W., Kung, H.-T., Liu, S., Deng, L., 2020b. Estimation of soil salt content using machine learning techniques based on remote-sensing fractional derivatives, a case study in the Ebinur Lake Wetland National Nature Reserve, Northwest China. Ecol. Indicators 119, 106869.
Wang, Y., Huang, T., Liu, J., Lin, Z., Li, S., Wang, R., Ge, Y., 2015. Soil pH value, organic matter and macronutrients contents prediction using optical diffuse reflectance spectroscopy. Comput. Electron. Agric. 111, 69–77.
Wang, Jiaqiang, Peng, Jie, Li, Hongyi, et al., 2021. Soil salinity mapping using machine learning algorithms with the sentinel-2 MSI in Arid Areas, China. Remote Sensing 13 (2), 305. https://doi.org/10.3390/rs13020305.
Wang, J., Shi, T., Yu, D., Teng, D., Ge, X., Zhang, Z., Yang, X., Wang, H., Wu, G., 2020a. Ensemble machine-learning-based framework for estimating total nitrogen concentration in water using drone-borne hyperspectral imagery of emergent plants: A case study in an arid oasis, NW China. Environ. Pollut. 266, 115412.
Wang, X., Zhang, F., Kung, H.-T., Johnson, V.C., 2018b. New methods for improving the remote sensing estimation of soil organic matter content (SOMC) in the Ebinur Lake Wetland National Nature Reserve (ELWNNR) in northwest China. Remote Sens. Environ. 218, 104–118.
Wang, L., Zhang, B., Shen, Q., Yao, Y., Zhang, S., Wei, H., Yao, R., Zhang, Y., 2021. Estimation of Soil Salt and Ion Contents Based on Hyperspectral Remote Sensing Data: A Case Study of Baidunzi Basin, China. Water 13.
Weng, Y., Gong, P., Zhu, Z., 2008. Soil salt content estimation in the Yellow River delta with satellite hyperspectral data. Can. J. Remote Sens. 34, 259–270.
Weng, Y.-L., Gong, P., Zhu, Z.-L., 2010. A Spectral Index for Estimating Soil Salinity in the Yellow River Delta Region of China Using EO-1 Hyperion Data. Pedosphere 20, 378–388.
Yang, N., Yang, S., Cui, W., Zhang, Z., Zhang, J., Chen, J., Ma, Y., Lao, C., Song, Z., Chen, Y., 2021. Effect of spring irrigation on soil salinity monitoring with UAV-borne multispectral sensor. Int. J. Remote Sens. 42, 8952–8978.
Zhang, B., Guo, B., Zou, B., Wei, W., Lei, Y., Li, T., 2022. Retrieving soil heavy metals concentrations based on GaoFen-5 hyperspectral satellite image at an opencast coal mine, Inner Mongolia, China. Environ. Pollut. 300, 118981.
Zhao, Q., Bai, J., Zhang, G., Jia, J., Wang, W., Wang, X., 2018. Effects of water and salinity regulation measures on soil carbon sequestration in coastal wetlands of the Yellow River Delta. Geoderma 319, 219–229.